Implementation Plan v1.0

February 2026 | Confidential

Thi do!ument brea" the Iron Tra!"er build into 12 e#uential mile tone $ ea!h de!ompo ed into individually te table tep . %very tep in!lude e&pli!it a!!eptan!e !riteria and a li t of te t that mu t pa before the tep i !on idered !omplete. ' tep are mall enough to !omplete in 1() day . * ile tone repre ent hippable in!rement that !an be demon trated to ta"eholder .

			!	
* 1	Pro+e!t ' !affolding , CI-C .	6	/ ee" 1	/ ee" 1
* 2	. ataba e ' !hema , 0uth	1	/ ee" 2	/ ee" 2
*)	%&er!i e 2ibrary 0PI , 3I	4	/ ee")	/ ee")
* 4	%#uipment 5ariant ' y tem	1	/ ee")(4	/ ee" 4
* 6	' et 2ogging Core	10	/ ee" 4(6	/ ee" 6
* 6	7e t Timer , 2ogging 3 8	6	/ ee" 6(6	/ ee" 6
* 4	' e ion 9rouping , : i tory	4	/ ee" 6(4	/ ee" 4
* 1	Progre Tra!"ing , P7	;	/ ee" 4(1	/ ee" 1
* ;	0nalyti! , 5i uali<ation	1	/ ee" ;(10	/ ee" 10
* 10	=ffline(Fir t , P / 0	4	/ ee" 10(11	/ ee" 11
* 11	0I * a!hine Identifi!ation	4	/ ee" 11(12	/ ee" 12
* 12	0I Coa!hing , 7e!ommendation	1	/ ee" 1)(16	/ ee" 16

" #

$ %& '( )*+ pyte t > pyte t(a yn!io. %very Fa t0PI endpoint$ Pydanti! model$ and ervi!e fun!tion. * o!" upaba e(py !all . Target? ;0@> line !overage on bu ine logi! module .

$ %,)*+ 5ite t > 7ea!t Te ting 2ibrary. %very !omponent render$ u er intera!tion$ and hoo" behavior. * o!" Tan ' ta!" Auery and ' upaba e !lient. Target? 16@> line !overage.

%& '( )*+ pyte t Bith a te t ' upaba e pro+e!t Cdedi!ated free(tier in tan!eD. 7eal . E operation $ 72 ' verifi!ation$ auth floB validation.

%,)*+ 5ite t Bith * ' / C* o!" ' ervi!e / or"erD inter!epting ' upaba e and Fa t0PI !all . Full u er floB Bithin a ingle !reen.

)- -) + PlayBright. Criti!al path ? ign up$ log a et$ vieB hi tory$ vieB !hart. 7un again t the Fetlify previeB deploy > 7ender taging > ' upaba e te t pro+e!t. Target? 16(20 %2% !enario !overing happy path and "ey error tate .

' + Pydanti! model e&port hared a Type ' !ript type via a !odegen tep Cpydanti!(to(type !ript or openapi(type !ript from Fa t0PIG =pen0PI pe!D. Frontend type are alBay in yn! Bith ba!"end !hema .

.

& '/ # + main Cprodu!tionD$ develop CintegrationD$ feature-H Cper(tep bran!he D. %a!h tep merge to develop via P7 Bith CI pa ing. * ile tone merge develop to main and trigger a produ!tion deploy.

, + Conventional Commit ? featC * 2(')D? implement 72 ' poli!ie for et table.

, #+ Ea!"end? te t -te t I JmoduleK.py. Frontend? JComponentK.te t.t & or Jhoo"K.te t.t ad+a!ent to the our!e file.

+ Code revieBed$ all te t pa ing in CI$ no regre ion in e&i ting te t $ a!!eptan!e !riteria verified manually or via %2% te t.

" 0 1 2 ' ') # 3 4 !

9oal? 0 deployable "eleton Bith empty !reen $ CI pipeline running lint and te t on every pu h$ and previeB deploy for every P7. Fo bu ine logi! yet.

- 5 & '( )

Create the Fa t0PI pro+e!t "eleton Bith a modular folder tru!ture$ pypro+e!t.toml for dependen!y management$ and a health endpoint.

" ' +

- ✓ 9%T -api-health return JL tatu L? Lo"LK Bith 200
- ✓ Pro+e!t u e Python).12>$ Fa t0PI 0.116>$ Pydanti! v2$ pydanti!(etting
- ✓ Folder tru!ture? ba!"end-app-Jauthaianalyti! $!ron$!oreK- ea!h Bith I I init I I .py
- ✓ ba!"end-app-!ore-!onfig.py u e pydanti!(etting Ea e ' etting Bith .env loading
- ✓ re#uirement .t&t or pypro+e!t.toml lo!"ed Bith all dep

+

- ▶ 3nit? te t I health I endpoint return 200 and !orre!t M ' = F body
- ▶ 3nit? te t I !onfig I load I from I env verifie Ea e ' etting read ' 3P0E0 ' %I 3 72$ ' 3P0E0 ' %I N%O$ 0FT : 7 = PIC I 0PI I N%O Bith default -validation
- ▶ 3nit? te t I !onfig I mi ing I re#uired I rai e rai e 5alidation%rror Bhen re#uired env var are ab ent
- ▶ 3nit? te t I !or I middleBare I !onfigured verifie C = 7 ' alloB !onfigured origin

- " 5 ,)

Create the 7ea!t > 5ite > Type ' !ript pro+e!t Bith Tan ' ta!" 7outer Cfile(ba ed D$ Tan ' ta!" Auery$ * 3I Bith a !u tom * .) theme$ and tub page for ea!h bottom nav tab.

" ' +

- ✓ 5ite dev erver tart Bithout error
- ✓ Tan ' ta!" 7outer file(ba ed routing !onfigured Bith route for -log$ -hi tory$ - tat $ -profile
- ✓ * 3I ThemeProvider Brap the app Bith dar"(mode(default * .) !olor to"en
- ✓ Eottom FavigationEar render four tab C2og$: i tory$ ' tat $ Profile D and navigate !orre!tly
- ✓ AueryClientProvider Brap the app Bith default taleTime and g!Time !onfiguration
- ✓ Type ' !ript tri!t mode enabled$ no type error

+

- ▶ 3nit? te t I app I render I Bithout I !ra h P render Q0pp -R and verifie bottom nav i pre ent
- ▶ 3nit? te t I bottom I nav I tab P verifie 4 tab render Bith !orre!t label C2og$: i tory$ ' tat $ Profile D
- ▶ 3nit? te t I navigation P !li!"ing ea!h tab navigate to the !orre!t route
- ▶ 3nit? te t I dar" I theme I default P verifie * 3I theme palette.mode SSS Tdar"T
- ▶ 3nit? te t I #uery I !lient I !onfigured P verifie AueryClient default are et
- ▶ ' nap hot? te t I theme I to"en P nap hot of the theme ob+e!t to !at!h a!!idental !hange

- . # 1 %& '( )*

' et up 9it : ub 0!tion Bor"floB for the ba!"end? in tall dep $ run pyte t$ run ruff linter$ run mypy type !he!"er on every pu h and P7.

" ' +

- ✓ / or"floB file at .github-Bor"floB -ba!"end(!i.yml
- ✓ 7un on pu h to develop and main$ and on all P7
- ✓ ' tep ? !he!"out$ etup Python).12$ in tall dep $ ruff !he!"$ mypy$ pyte t Bith !overage report
- ✓ Fail the build on any lint error$ type error$ or te t failure
- ✓ Coverage report uploaded a artifa!t

  # +

- ▶ CI * eta? pu h a deliberately failing te t and verify CI report failure
- ▶ CI * eta? pu h a ruff violation and verify CI report failure
- ▶ CI * eta? pu h !lean !ode and verify CI report u!!e Bith !overage artifa!t

- 6 # 1 %,)*

' et up 9it : ub 0!tion Bor"floB for the frontend? in tall dep $ run 5ite t$ run % ' 2int$ run Type ' !ript type !he!" on every pu h and P7.

" ' +

- ✓ / or"floB file at .github-Bor"floB -frontend(!i.yml
- ✓ 7un on pu h to develop and main$ and on all P7
- ✓ ' tep ? !he!"out$ etup Fode 20$ npm !i$ e lint$ t ! ((no%mit$ vite t run Bith !overage
- ✓ Fail the build on any lint error$ type error$ or te t failure
- ✓ Coverage report uploaded a artifa!t

  # +

- ▶ CI * eta? pu h a deliberately failing te t and verify CI report failure
- ▶ CI * eta? pu h an % ' 2int violation and verify CI report failure
- ▶ CI * eta? pu h !lean !ode and verify CI report u!!e

- 7 3) !

Configure Fetlify for frontend previeB deploy on P7 and produ!tion deploy on main. Configure 7ender for the ba!"end Bith environment variable .

" ' +

- ✓ Fetlify ite !onne!ted to the repo Bith build !ommand? !d frontend , , npm !i , , npm run build
- ✓ Fetlify publi h dire!tory? frontend-di t
- ✓ Fetlify previeB deploy generate a 372 for every P7
- ✓ 7ender Beb ervi!e !onne!ted to the repo Bith root dire!tory? ba!"end
- ✓ 7ender tart !ommand? uvi!orn app.main?app ((ho t 0.0.0.0 ((port UP=7T
- ✓ %nvironment variable !onfigured on both platform ℂno e!ret in repo℗
- ✓ C=7 ' !onfigured on ba!"end to alloB Fetlify previeB 372 and produ!tion 372

  # +

- ▶ ' mo"e? deploy to Fetlify previeB$ verify the app load and bottom nav render

▶ ' mo"e? deploy to 7ender$ verify 9%T -api-health return 200

▶ ' mo"e? verify C=7 ' P frontend previeB !an !all ba!"end health endpoint Bithout C=7 ' error

- 8 1 2 '

Create the ' upaba e pro+e!t$!onfigure auth provider $ and verify !onne!tivity from both frontend and ba!"end.

" ' +

✓ ' upaba e pro+e!t !reated Bith Po tgre ' A2 16

✓ %mail auth provider enabled

✓ 9oogle =0uth provider !onfigured C!lient I . - e!retD

✓ ' upaba e 372 and anon "ey added to frontend env

✓ ' upaba e 372 and ervi!e role "ey added to ba!"end env

✓ V upaba e- upaba e(+ initiali<ed in frontend Bith !reateClientCD

✓ upaba e(py a yn! !lient initiali<ed in ba!"end life pan event

+

▶ Integration? te tI upaba eIfrontendI!onne!tion P frontend !an !all upaba e.auth.get ' e ionCD Bithout error

▶ Integration? te tI upaba eIba!"endI!onne!tion P ba!"end !an !all upaba e.tableCTIte tIpingTD. ele!tCTHTD C!reate and drop a temp te t tableD

▶ Integration? te tIauthI ignupIfloB P !reate a te t u er via ' upaba e 0uth$ verify the u er e&i t $ delete the u er

. "0! '/ 3 /

9oal? 0ll !ore table !reated in 'upaba e Bith 72' poli!ie $a Bor"ing auth floB C ignup$ login$ logout$ to"en refre hD end(to(end$ and the Fa t0PI M / T verifi!ation dependen!y.

"- 9 ' 3)!

Create the e&er!i e table for the !anoni!al e&er!i e library. Populate Bith W400 eed e&er!i e !overing all ma+or mu !le group and e#uipment type . %&er!i e are publi!ly readable but only admin Cor the y temD !an in ert eed data.

" ' +

- ✓ Table e&er!i e !reated Bith !olumn ?id Cuuid PND$ name$!ategory$ primaryImu !le Cte&tXYD$ e!ondaryImu !le Cte&tXYD$ movementItype$ i I!u tom$!reatedIby Cnullable FN to auth.u er D$!reatedIat
- ✓ 72' enabled? '%2%CT alloBed for all authenti!ated u er $IF '%7T-3P . 0T%- . %2%T% re tri!ted to roB Bhere !reatedIby S auth.uidCD C!u tom e&er!i e D or via ervi!e role C eed dataD
- ✓ 'eed 'A2 file in ert W400 e&er!i e Bith !orre!t mu !le group mapping
- ✓ Inde&e on? name Cfor ear!hD$!ategory$ primaryImu !le C9IF for array !ontainmentD

+

- ▶ Integration? te tIe&er!i e IreadableIbyIauthenti!ated P authenti!ated u er !an '%2%CT all eeded e&er!i e
- ▶ Integration? te tIe&er!i e InotIreadableIbyIanon P unauthenti!ated re#ue t return 0 roB or 401
- ▶ Integration? te tI eedIdataI!ount P verify Z400 e&er!i e e&i t after eed
- ▶ Integration? te tI eedI!over IallI!categorie P verify e&er!i e e&i t for pu h$ pull$ leg $!ore$!ardio
- ▶ Integration? te tI ear!hIbyIname P '%2%CT / :%7% name I2IN%T@ben!h@T return e&pe!ted re ult
- ▶ Integration? te tIu erI!annotImodifyI eed P 3P . 0T%- . %2%T% on eed e&er!i e C!reatedIby I' F322D fail for normal u er

"-" : ;

Create the e#uipmentIvariant table that tore ma!hine(pe!ifi! in tan!e of e&er!i e $per u er.

" ' +

- ✓ Table e#uipmentIvariant Bith !olumn ?id Cuuid PND$ u erIid CFND$ e&er!i eIid CFND$ name$ e#uipmentItype$ manufa!turer$ BeightIin!rement$ BeightIunit$ eatI etting C+ onbD$ note $ photoIurl$ i Idefault$ la tIu edIat$!reatedIat
- ✓ 72'? all operation re tri!ted to roB Bhere u erIid S auth.uidCD
- ✓ Foreign "ey to e&er!i e CidD Bith =F . %2%T% C0 'C0 . %
- ✓ 3ni#ue !on traint on Cu erIid$ e&er!i eIid$ nameD to prevent dupli!ate variant name per u er per e&er!i e
- ✓ Inde& on Cu erIid$ e&er!i eIid$ la tIu edIat . %'CD for *73 orting

+

▶ Integration? te tl!reatelvariant P authenti!ated u er !an IF '%7T a variant lin"ed to an e&er!i e

▶ Integration? te tlvariantli olation P u er 0 !annot '%2%CT u er E variant

▶ Integration? te tlvariantlupdate P oBner !an 3P . 0T% their variant name and eatl etting

▶ Integration? te tlvariantldelete P oBner !an . %2%T% their variant

▶ Integration? te tldupli!atelnamelre+e!ted P IF '%7T ame Cu erlid$ e&er!i elid$ nameD return uni#ue !on traint error

▶ Integration? te tl!a !adelonle&er!i eldelete P deleting a !u tom e&er!i e !a !ade to it variant

▶ Integration? te tl+ onbl eatl etting P IF '%7T- '%2%CT Bith !omple& M' =FE CJ eatlheight?)$ ba!"lpad? 2KD Bor" !orre!tly

"- .

Create the et table P the atomi! unit of training data. Thi i the mo t !riti!al table for performan!e and mu t upport rapid in ert and time(range #uerie .

" ' +

✓ Table et Bith !olumn ? id Cuuid PND$ u erlid CFND$ e&er!i elid CFND$ variantlid CFN nullableD$ Beight Cde!imalD$ Beightlunit$ rep CintD$ rpe Cde!imal nullableD$ rir Cint nullableD$ etltype$ re tl e!ond Cint nullableD$ durationl e!ond Cint nullableD$ tempo Cte&t nullableD$ note Cte&t nullableD$ loggedlat Ctime tampt<D$!reatedlat$ e ionlgroup Cdate$ generated alBay a . 0T%Clloggedlat DD

✓ 72'? all operation re tri!ted to u erlid S auth.uidCD

✓ Foreign "ey to e&er!i e CidD and e#uipmentlvariant CidD$ both Bith =F . %2%T% '%T F322 for variantlid

✓ Inde&e ? Cu erlid$ loggedlat . %'CD$ Cu erlid$ e&er!i elid$ loggedlat . %'CD$ Cu erlid$ e ionlgroupD

✓ C : %CN !on traint ? Beight RS 0$ rep RS 0$ rpe E%T / %%F 6.0 0F . 10.0 =7 F322$ rir E%T / %%F 0 0F . 6 =7 F322

+

▶ Integration? te tlin ertl et P authenti!ated u er !an IF '%7T a et Bith re#uired field

▶ Integration? te tlin ertl etlBithlvariant P et lin"ed to a valid variantlid u!!eed

▶ Integration? te tlin ertl etlBithoutlvariant P et Bith variantlid S F322 u!!eed Cgeneri! loggingD

▶ Integration? te tl etli olation P u er 0 !annot '%2%CT u er E et

▶ Integration? te tl etl#uerylbyle&er!i e P '%2%CT / : %7% e&er!i elid S 8 =7 . %7 EO loggedlat . %'C return !orre!t re ult

▶ Integration? te tl etl#uerylbyl e ionlgroup P '%2%CT / : %7% e ionlgroup S T2026(02(24T return all et from that date

▶ Integration? te tl!he!"l!on traintlBeight P IF '%7T Bith Beight S (6 fail

▶ Integration? te tl!he!"l!on traintlrpe P IF '%7T Bith rpe S 11 fail $ rpe S 1.6 u!!eed

▶ Integration? te tl e ionlgroublgenerated P IF '%7T a et at 2026(02(24T2)?)0?00[$ verify e ionlgroup S 2026(02(24

▶ Integration? te tlvariantlf"l etlnull P . %2%T% variant$ verify et variantlid be!ome F322 Cnot !a !ade deleteD

▶ Performan!e? te tlin ertllaten!y P IF '%7T 100 et in e#uen!e$ verify avg laten!y Q 60m

"- 6 ; ') 3

Create the per onalIre!ord table and a Po tgre ' A2 trigger fun!tion that !he!" for neB P7 on every et in ert.

" ' +

✓ Table per onalIre!ord Bith !olumn ? id Cuuid PND$ u erIid CFND$ e&er!i eIid CFND$ variantIid CFN nullableD$ re!ordItype Cte&t? e tI1rm$ repIma&I1$ repIma&I)$ repIma&I6$ repIma&I1$ repIma&I10$ ma&IvolumeD$ value CdeIimalD$ etIid CFN to et D$ a!hievedIat Ctime tampt<D$!reatedIat

✓ 72 '? '%2%CT- . %2%T% for u erIid S auth.uidCD$ IF '%7T-3P . 0T% via trigger only C ervi!e roleD

✓ 3ni#ue !on traint on Cu erIid$ e&er!i eIid$ variantIid$ re!ordItypeD for up ert emanti!

✓ Trigger fun!tion !he!"Iper onalIre!ordCD fire 0FT%7 IF '%7T on et ? !ompute e tI1rm via %pley$!ompare again t e&i ting P7$ up ert if beaten

✓ Trigger al o !he!" repIma& re!ord ? e.g.$ if rep S6 and BeightR!urrent 67 * re!ord

+

▶ Integration? te tIfir tI etI!reate Ipr P IF '%7T a et for a neB e&er!i e$ verify per onalIre!ord roB !reated for e tI1rm and repIma&

▶ Integration? te tIbetterI etIupdate Ipr P IF '%7T a et Bith higher e tI1rm than e&i ting$ verify P7 updated Bith neB etIid

▶ Integration? te tIBor eI etInoIprI!hange P IF '%7T a et Bith loBer e tI1rm$ verify P7 un!hanged

▶ Integration? te tIrepIma&Itra!"ing P IF '%7T et at 100"g&6$ then 106"g&6$ verify repIma&I6 updated to 106

▶ Integration? te tIvolumeIre!ord P IF '%7T et at 100"g&10 CvolS1000D$ then 10"g&16 CvolS1200D$ verify ma&Ivolume updated

▶ Integration? te tIvariantI pe!ifi!Ipr P P7 for variant 0 doe not affe!t P7 for variant E of the ame e&er!i e

▶ 3nit? te tIepleyIformula P verify e17 * !al!ulation? 100"g & 1 rep S 100 H C1 > 1-)OD S 126.64

▶ 3nit? te tIbr<y!"ilformula P verify fallba!" for R12 rep ? Beight H)6 - C)4 (rep D

"- 7 ;

Create the orene Ireport table for po t(Bor"out mu !le orene tra!"ing C!arried forBard from vOD.

" ' +

✓ Table orene Ireport Bith !olumn ? id Cuuid PND$ u erIid CFND$ mu !leIgroup Cte&tD$ orene Ilevel Cint 0(4D$ reportedIat Ctime tampt<D$ e ionIgroup Cdate$ referen!ing Bhi!h Bor"out triggered the promptD$!reatedIat

✓ 72 '? all operation re tri!ted to u erIid S auth.uidCD

✓ C : %CN !on traint? orene Ilevel E%T / %%F 0 0F . 4

+

▶ Integration? te tIin ertI orene P authenti!ated u er !an report orene for T!he tT at level)

▶ Integration? te tI orene Ii olation P u er 0 !annot ee u er EG report

▶ Integration? te tIinvalidIlevelIre÷e!ted P IF '%7T Bith orene Ilevel S 6 fail

▶ Integration? te tI#ueryIre!entI orene P '%2%CT / : %7% u erIid S 8 0F . reportedIat R noBCD (intervalT4 day T return re!ent report

"- 8 ';' '/

Create the analyti! l!a!he table for erver(ide pre(!omputed analyti! CM' =FE payload Bith type di !riminatorD.

" ' +

✓ Table analyti! l!a!he Bith !olumn ?id Cuuid PND$ u erlid CFND$!a!heltype Cte&t? Bee"lylvolume$ e&er!i el1rmltrend$ mu !leldi tribution$ trainglfre#uen!y$ trengthl !oreD$ payload C+ onbD$ periodl tart CdateD$ periodlend CdateD$!omputedlat Ctime tampt<D$!reatedlat

✓ 72'? '%2%CT for u erlid S auth.uidCD$ IF '%7T-3P . 0T%- . %2%T% via ervi!e role only Cba!"end !ronD

✓ 3ni#ue !on traint on Cu erlid$!a!heltype$ periodl tart$ periodlendD

✓ Inde& on Cu erlid$!a!heltype$ periodlend . %' CD

+

▶ Integration?te tl!a!helin ertlvial ervi!elrole P ervi!e role !an IF '%7T analyti! !a!he entry

▶ Integration?te tl!a!helreadablelbylloBner P authenti!ated u er !an '%2%CT their oBn !a!he entrie

▶ Integration?te tl!a!hellnotlBritablelbylu er P normal authenti!ated u er !annot IF '%7T-3P . 0T% !a!he entrie

▶ Integration?te tl!a!hellup ert P e!ond IF '%7T Bith ame uni#ue "ey up ert C=F C=FF2ICT 3P . 0T%D

"- < , 1 => ? '

Create the getl!urrentlu er dependen!y that verifie 'upaba e M/T u ing M/N' Ca ymmetri! igningD. Thi dependen!y Bill prote!t all authenti!ated ba!"end endpoint .

" ' +

✓ . ependen!y fun!tion getl!urrentlu erCauthori<ation? tr S :eaderCDD e&tra!t Eearer to"en

✓ Fet!he 'upaba e M/N' from J'3P0E0'%l372K-auth-v1-.Bell("noBn-+B" .+ on C!a!hed for 1 hourD

✓ 5erifie M/T ignature$ e&piry$ and i uer u ing python(+o e or PyM/T

✓ %&tra!t u erlid from the ub !laim

✓ 7eturn a 3 erConte&tCu erlidSuuid$ emailS trD data!la

✓ 7ai e 401 on invalid-e&pired to"en$ 40) on mi ing to"en

+

▶ 3nit?te tlvallidl+Btlreturn lu er P mo!" M/N'$!reate a valid M/T$ verify 3 erConte&t returned Bith !orre!t u erlid

▶ 3nit?te tle&piredl+Btlrai e l401 P mo!" M/N'$!reate an e&pired M/T$ verify :TTP%&!eptionC401D

▶ 3nit?te tlinvalidl ignaturelrai e l401 P ign M/T Bith Brong "ey$ verify :TTP%&!eptionC401D

▶ 3nit?te tlmi inglauthlheaderlrai e l40) P !all Bithout 0uthori<ation header$ verify :TTP%&!eptionC40)D

▶ 3nit?te tlmalformedlto"enlrai e l401 P pa TEearer garbageT header$ verify :TTP%&!eptionC401D

▶ 3nit?te tl+B" l!a!hing P tBo !all Bithin 1 hour ma"e only one :TTP re#ue t to M/N' endpoint

▶ Integration?te tlfullauthlfloB P ign up via 'upaba e$ get a!!e to"en$!all a prote!ted endpoint$ verify 200

"- @ ,) / ,

Euild the login$ ignup$ and e ion management 3I u ing V upaba e- upaba e(+ $ Bith a 7ea!t !onte&t for auth tate and a Prote!ted7oute Brapper.

" ' +

- ✓ 0uthProvider !onte&t Brap the app$ e&po e ? u er$ e ion$ loading$ ign3p$ ignIn$ ign=ut
- ✓ 0uthProvider li ten to on0uth ' tateChange and update tate rea!tively
- ✓ 2ogin !reen Bith email-pa Bord field and a 9oogle =0uth button
- ✓ ' ignup !reen Bith email$ pa Bord$!onfirm pa Bord field
- ✓ Prote!ted7oute !omponent redire!t to -login if no e ion
- ✓ 0ll four main tab Brapped in Prote!ted7oute
- ✓ 0!!e to"en atta!hed to ' upaba e !lient automati!ally C upaba e(+ handle thi D
- ✓ To"en refre h handled automati!ally by upaba e(+

+

▶ 3nit? te tIauthIproviderIinitialI tate P render Bith loadingStrue$ then re olve to u erSnull if no e ion

▶ 3nit? te tI ignIupI!all I upaba e P mo!" upaba e.auth. ign3p$ verify it$!alled Bith !orre!t param

▶ 3nit? te tI ignIinI!all I upaba e P mo!" upaba e.auth. ignIn / ithPa Bord$ verify !orre!t param

▶ 3nit? te tI ignIout P mo!" upaba e.auth. ign=ut$ verify u er tate be!ome null

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